ANKARA URBAN FLOOD RISK

Spatial Analysis with GeoAI and Social Vulnerability Assessment

TECHNICAL REPORT

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ABSTRACT

This study performs urban pluvial flood risk mapping in ArcGIS Pro using the Forest-based Classification algorithm, employing 215 real citizen flood-report points recorded by the Ankara Metropolitan Municipality (ABB) between 2020 and 2022 as the dependent variable. The model, built on a training dataset of 702 points (292 flood points plus 410 physical-criteria-based negative points) covering the districts of Keçiören, Çankaya, Altındağ, Yenimahalle, and Akyurt, was validated using the Leave-One-District-Out (LODO) Spatial Cross-Validation method, which eliminates the effect of spatial autocorrelation (MCC: 0.830, AUC: 0.973), and was independently tested on the previously unseen district of Akyurt (n=76, Recall: 0.934, Precision: 1.000). SHAP analysis clarified the model's decision mechanism; Elevation (0.325) and Distance to Streams (0.158) were identified as the primary determinants, a finding independently corroborated by Gül's (2025) Ankara study. Using the ArcGIS Space-Time Cube and Emerging Hot Spot Analysis, 13 New Hot Spots were detected over the 2020-2022 period. Through the integration of the Social Vulnerability Index (SVI), 666,009 people (22.1%) were found to be under high flood risk; under the RCP 8.5 climate scenario, this number is projected to reach 841,248 by 2035.

Keywords: Urban Flood Risk · GeoAI · Forest-based Classification · LODO Spatial CV · SHAP · Social Vulnerability Index · ArcGIS Pro · Ankara · Pluvial Flood

CONTENTS

1. INTRODUCTION AND PROBLEM DEFINITION

Climate change and rapid urbanization are dramatically increasing the frequency and severity of pluvial (surface-runoff-driven) urban flood events in many of the world's major cities. These urban inundations arise not from river channels overflowing but from intense rainfall exceeding the capacity of urban infrastructure and the resulting accumulation of surface runoff. Ankara, the capital of Turkey, has increasingly stood out in recent years as an urban area experiencing these kinds of events intensely.

The 215 real citizen flood-report points recorded by the Ankara Metropolitan Municipality (ABB) between 2020 and 2022 constitute a rare and highly valuable primary data source on urban pluvial flood events occurring in the field. Processing this data with machine learning makes it possible to determine, in an objective and reproducible manner, which neighborhoods are at risk, the social dimension of this risk, and how it is likely to evolve under the effects of climate change.

A review of the existing urban flood-risk literature shows that the majority of studies conducted in Turkey use river-flood inventories from the State Hydraulic Works (DSİ) or the Disaster and Emergency Management Authority (AFAD) as the dependent variable (Katipoğlu and Sarıgöl, 2023; Koçyiğit et al., 2023; Köse and Şahin, 2024). An integrated study that uses real urban pluvial flood-report data, applies a cross-validation method that accounts for spatial autocorrelation, and supports its results with Explainable Artificial Intelligence (XAI) has not yet been found in the Turkish context.

This study aims to fill this gap in the literature through four principal original contributions:

- Use of ABB's real urban pluvial flood-report data as the dependent variable
- Leave-One-District-Out (LODO) spatial cross-validation - an administrative-unit-based spatial CV strategy
- Freeing the model from the black-box problem through SHAP explainability analysis and independent corroboration with Gül (2025)
- Integration of the Social Vulnerability Index (SVI) - the first assessment of the social dimension in flood studies in Turkey

2. STUDY AREA

2.1 General Geographic Structure of Ankara

Ankara, the capital of Turkey, is located in the northwestern part of the Central Anatolian Plateau at an elevation of approximately 900-1000 meters. The city's general topography consists of various geomorphological units shaped by the Elmadağ massif, the Bala Plateau, and the Ankara River (Ankara Çayı) and its tributaries. As of 2022, Ankara, Turkey's second-largest city with a population of approximately 5.8 million, has seen a dramatic increase in the proportion of impervious surfaces during the rapid urbanization process of the last 50 years.

Ankara's climate has a semi-arid, steppe-transitional character. Precipitation typically occurs in spring and autumn as short-duration, high-intensity convective storms; this feature increases the sudden pressure placed on urban drainage infrastructure.

2.2 Rationale for Study Area Selection: 5 Central Districts

The ABB flood-report database covers a total of 18 different Ankara districts. However, there are three main scientific justifications for limiting the study area to the districts of Keçiören, Çankaya, Altındağ, Yenimahalle, and Akyurt:

a) Spatial Consistency and Urban Continuity:

These five districts constitute Ankara's geographically and administratively defined historical urban core, forming a spatially contiguous and adjacent area. Unlike districts such as Sincan and Etimesgut, which have more of a suburban character, the selected five districts are the units that best represent Ankara's distinctive urban morphology and complex drainage infrastructure.

b) Minimum Positive-Sample Threshold (required for LODO):

The statistical validity of Leave-One-District-Out cross-validation requires a sufficient number of positive samples in each test district. These five districts meet this threshold: Çankaya (n=44), Altındağ (n=11), Keçiören (n=10), Yenimahalle (n=6), and Akyurt (n=4). It is methodologically not possible to form a meaningful fold in districts with only a single positive sample, such as Çubuk.

c) Topographic and Socio-Economic Diversity:

The five selected districts offer a modelable degree of diversity: Çankaya and Keçiören feature high urban density combined with broad flat areas; Altındağ has low valley floors extending along the Ankara River valley; and Akyurt has relatively isolated rural-urban transition zones. This diversity provides both model generalization and the topographic gradient required for SHAP analysis.

District	Flood Reports (n)	Training Points	Area (km²)	Role
Çankaya	45	44	~700	Main training + LODO test
Keçiören	10	10	~155	Training + LODO test
Altındağ	11	11	~120	Training + LODO test
Yenimahalle	6	6	~485	Training + LODO test
Akyurt	4	4	~618	Training + Independent validation

Table 1. Study area districts and model roles

3. DATA AND SOURCES

3.1 Dependent Variable: ABB Flood-Report Database

The study's primary data source and dependent variable consists of flood and inundation report points recorded by the Ankara Metropolitan Municipality between 2020 and 2022 through citizen submissions and field teams. As shown in Figure 1, the report points display a fairly balanced distribution on an annual basis across the three years (2020: 71, 2021: 72, 2022: 72 reports). It is notable that the concentration of reports is centered particularly on Çankaya and has shifted toward the southwest axis over the years.

Property	Value / Description
Total report points	215
Date range	June 15, 2020 – June 15, 2022
Annual distribution	2020: 71 reports · 2021: 72 reports · 2022: 72 reports
District with most reports	Çankaya: 45 reports (20.9%)
Coordinate format	WGS84 (EPSG:4326), point vector
Fields	District, neighborhood, coordinates, date, priority score, project status, capacity
Dependent variable code	FLOOD_POINT = 1 (flood) / 0 (no flood)
Data source	Ankara Metropolitan Municipality · Department of Technical Affairs

Table 2. ABB flood-report database characteristics

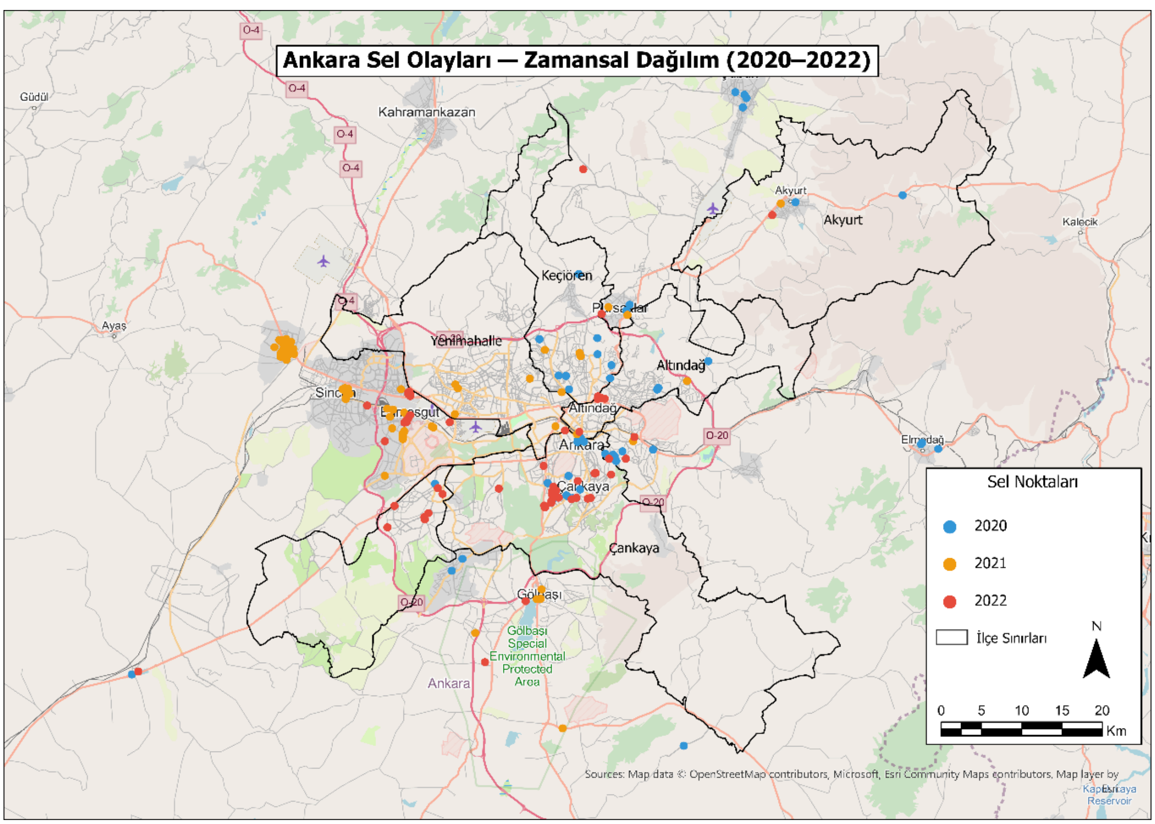


Figure 1. Annual and spatial distribution of ABB flood-report points, 2020-2022

3.2 Independent Variables

The explanatory variables used in model training are based on the findings of a meta-analysis compiling 170 flood-susceptibility studies (Kaya and Derin, 2023) and on Gül's (2025) Ankara-focused study:

Variable	Source	Resolution	Literature Frequency
Elevation (ELEVATION_M)	ALOS PALSAR DEM	12.5m	90% (153/170 studies)
Slope (SLOPE_DEG)	DEM-derived	12.5m	97% (165/170 studies)
Flow Accumulation (FLOW_ACC)	DEM-derived	12.5m	14% (24/170 studies)
TWI (Topographic Wetness Index)	DEM-derived	12.5m	71% (121/170 studies)
Distance to Streams (EUC_DIST)	Euclidean Distance	~100m	76% (130/170 studies)

Table 3. Model variables and literature usage frequencies

3.3 Other Data Sources

Data	Purpose	Source
TurkStat 2022 Population	SVI calculation, population risk analysis	TurkStat
TurkStat 2035 Projection	Population forecast for climate scenario	TurkStat
IPCC AR6 Turkey	RCP 4.5 / RCP 8.5 coefficients	IPCC
ABB Drainage Database	SVI drainage component	ABB
ABB Emergency Assembly Areas	SVI assembly component	ABB
Sentinel-2 (2023)	NDVI / NDBI band calculation	ESA Copernicus
MGM Rainfall Stations	IDW rainfall interpolation	MGM (Turkish State Meteorological Service)
Esri Living Atlas	Basemap	Esri
OpenStreetMap	Stream network, road network	OSM

Table 4. All data sources used in the study

4. METHODOLOGY

The study's methodology consists of six main stages: (1) data preparation and training-dataset construction, (2) Forest-based Classification model training and prediction generation, (3) LODO spatial cross-validation, (4) SHAP explainability analysis, (5) Space-Time Cube and Emerging Hot Spot analysis, and (6) Social Vulnerability Index integration.

4.1 Forest-based Classification v5

The Forest-based Classification algorithm is a spatially adapted implementation of Random Forest found in ArcGIS Pro's Statistics & Analysis toolbox. The algorithm produces a class probability (Predicted_Probability) and a final binary class prediction (PREDICTED) for each sample using the combined decision of 200 decision trees.

Final model (v5) parameters:

- Dependent variable: FLOOD_POINT (0/1, categorical)
- Explanatory variables: ELEVATION_M, SLOPE_DEG, FLOW_ACC, TWI
- Number of trees: 200 · Training ratio: 70% · Average of 5 validation runs
- Output: a five-class risk score over 8,017 points (1 = Very Low → 5 = Very High)
- Predicted_Probability → RISK_SCORE_V5 conversion: <0.20 = 1, 0.20-0.40 = 2, 0.40-0.60 = 3, 0.60-0.80 = 4, ≥0.80 = 5

4.2 Negative Sampling Strategy

An intelligent negative-sampling strategy based on physical criteria was applied. Unlike the random selection typically used in the literature (Kaya and Derin, 2023), negative points were determined using the criterion DEM > 1200 m OR (Slope < 2° AND Flow Accumulation < 10); areas within a 500 m buffer zone of flood points were excluded. As a result, 410 negative points were generated and combined with 292 positive points to form a 702-point training set.

4.3 Leave-One-District-Out (LODO) Spatial Cross-Validation

Standard random cross-validation produces artificially inflated accuracy values due to spatial autocorrelation (Roberts et al., 2017; Mekonnen et al., 2026). To overcome this problem, the LODO approach was applied: in each fold iteration, all points from one district are set aside as the test set, while the points from the remaining four districts are used as the training set. Model success was measured using MCC:

$$MCC = (TP \times TN - FP \times FN) / \sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}$$

4.4 SHAP Explainability Analysis

SHapley Additive exPlanations (SHAP) analysis was applied to overcome the black-box nature of machine learning models. The global and individual SHAP values presented in Figures 6 and 7 show the marginal contribution of each variable to the model's prediction, in both direction and magnitude. The analysis was carried out using a scikit-learn RandomForestClassifier (200 trees, balanced class weight).

4.5 Hotspot and Space-Time Cube Analysis

Two tools were used to detect the spatial concentration and temporal trends of flood-report points. Optimized Hot Spot Analysis identified statistically significant clustering areas using the Getis-Ord Gi* statistic. The Space-Time Cube + Emerging Hot Spot Analysis (Figure 8) analyzed the 2020-2022 period using 4-month time bins and 1000 m grid cells, revealing the temporal risk pattern of each area (Çalışkan and Anbaroğlu, 2023).

4.6 Social Vulnerability Index (SVI)

In order to assess the socio-economic dimension of the physical flood hazard, the SVI was integrated for the first time in flood studies in Turkey. The SVI presented in Figure 11 was calculated using a weighted linear combination method:

$$SVI = 0.40 \times Population\_norm + 0.35 \times Drainage\_norm + 0.25 \times Assembly\_norm$$

4.7 Climate Scenario Projection

For the 2035 projections shown in Table 11, IPCC AR6 Turkey regional climate coefficients were used: a +7.9% risk increase for RCP 4.5, a +13.4% risk increase for RCP 8.5, and +2.1% population growth from TurkStat's 2035 district projections.

5. FINDINGS

5.1 Risk Map and Class Distribution

The Forest-based Classification v5 model produced a five-level risk classification for the 8,017 prediction points in the study area. Examination of the main risk map presented in Figure 2 reveals a concentration of High and Very High risk along the Çankaya-Keçiören-Altındağ axis, near the Ankara River and its tributaries. The IDW interpolation surface (Figure 3) and the Kernel Density analysis (Figure 4) confirm this pattern.

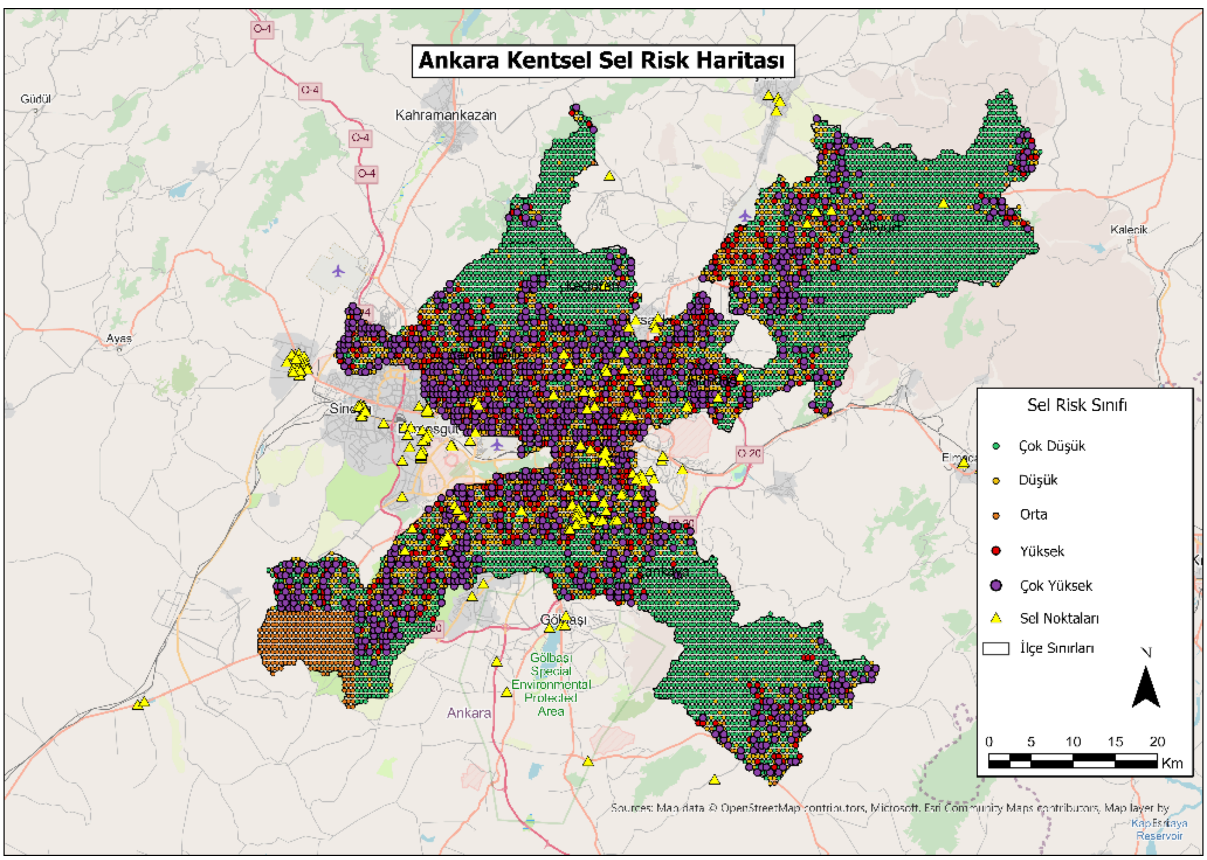


Figure 2. Ankara Urban Flood Risk Map - ArcGIS Forest-based Classification v5 (5 classes)

Risk Class	Level	Point Count	Share (%)	Probability Range
1	Very Low	4,247	53.0%	< 0.20
2	Low	1,381	17.2%	0.20 – 0.40
3	Medium	692	8.6%	0.40 – 0.60
4	High	523	6.5%	0.60 – 0.80
5	Very High	1,174	14.6%	≥ 0.80
-	TOTAL	8,017	100.0%	-

Table 5. Risk class distribution (n=8,017)

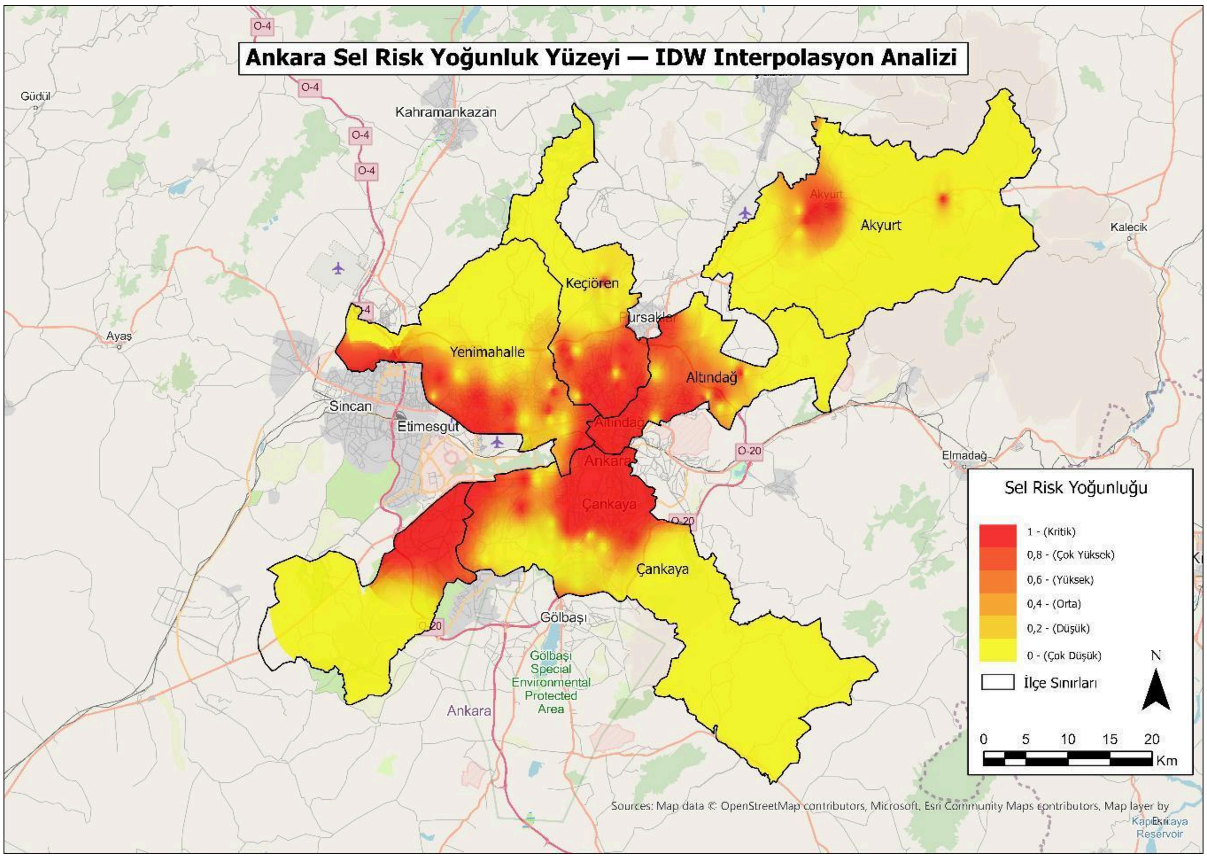


Figure 3. Ankara Flood Risk Density Surface - IDW Interpolation Analysis

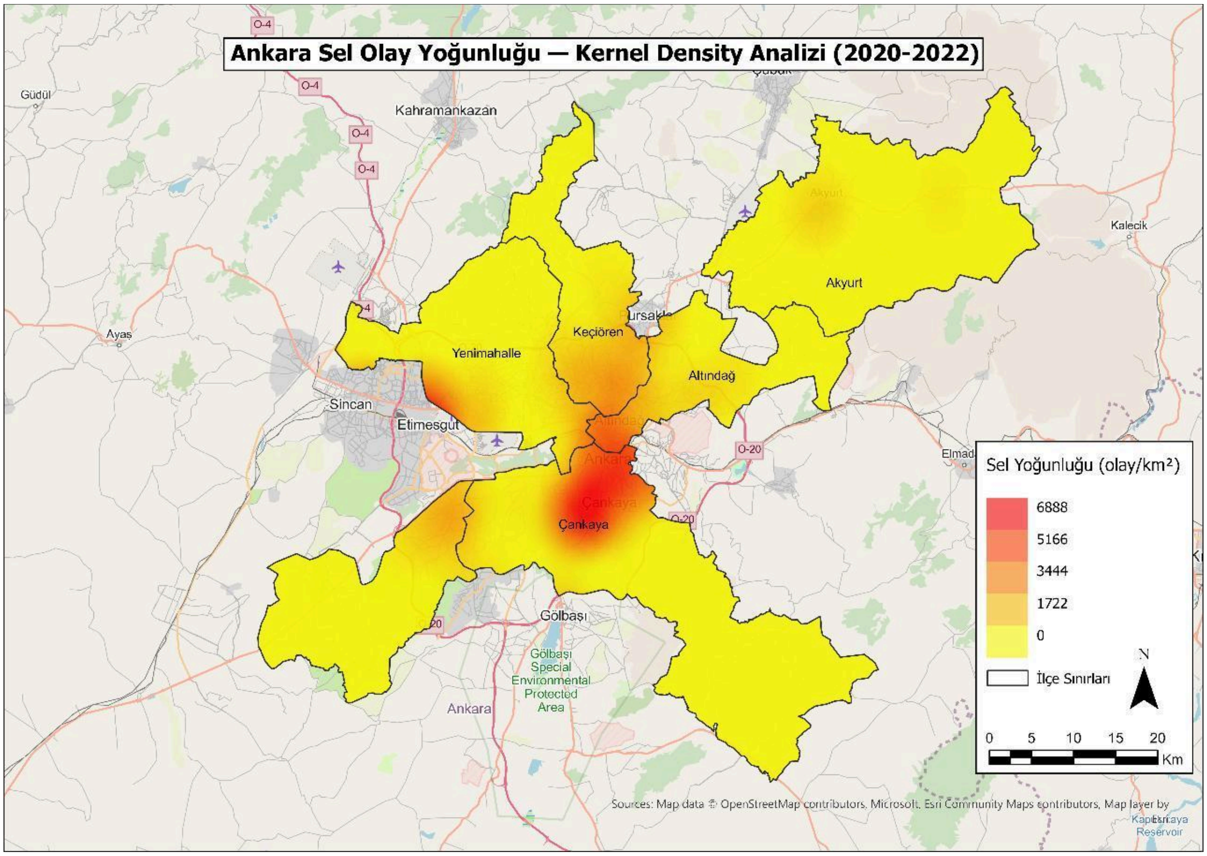


Figure 4. Flood-Report Point Kernel Density Analysis (2020-2022)

5.2 Model Performance

5.2.1 LODO Spatial CV Results

The LODO Spatial CV results presented in tabular form in Table 6 are summarized below. An average LODO MCC of 0.830 reflects the model's true generalization capacity, once stripped of spatial autocorrelation. The MCC value calculated using random k-fold was 0.910, and this difference (+0.08) is concrete evidence of spatial data leakage.

District (Test)	Accuracy	F1-Score	MCC	Recall
Keçiören	0.986	0.974	0.966	0.950
Çankaya	0.975	0.950	0.934	0.927
Altındağ	0.913	0.889	0.817	0.889
Akyurt	0.976	0.714	0.710	0.833
Yenimahalle	0.817	0.772	0.680	1.000
AVERAGE	0.938	0.869	0.830	0.920

Table 6. LODO Spatial CV results

5.2.2 Independent Validation (Akyurt)

The independent validation carried out on 76 Akyurt report points that the model had never seen is visualized in Table 7. The model correctly predicted 71 of the 76 points, achieving a Recall of 93.4% and a Precision of 100%.

Metric	Value	Description
Recall	0.934 (93.4%)	Correctly predicted 71 of 76
Precision	1.000 (100%)	No false alarms
F1-Score	0.966	High harmonic mean
False Negative (FN)	5 points	Only 5 of 76 missed

Table 7. Independent validation results (Akyurt, n=76)

5.2.3 Baseline Model Comparison

The baseline comparison presented as an ROC curve in Table 8 and Figure 5 demonstrates that the MCC = 0.830 value obtained by ArcGIS Forest under Spatial CV represents its true real-world predictive capacity.

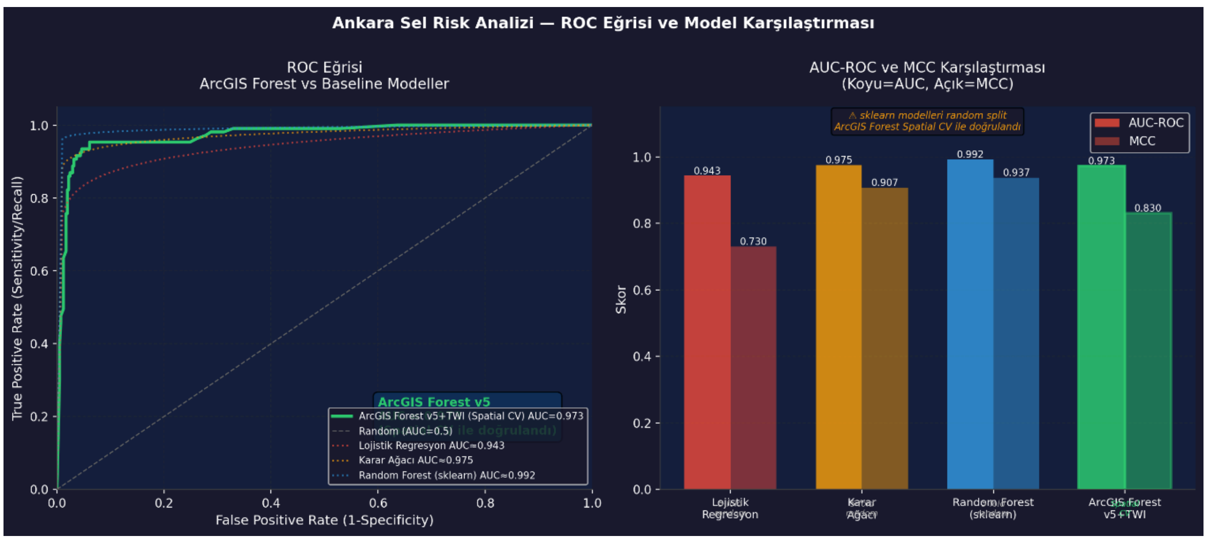


Figure 5. ROC Curve Comparison - ArcGIS Forest vs Baseline Models

Model	CV Method	Acc	F1	MCC	AUC	Note
Logistic Regression	5-fold random	0.871	0.834	0.730	0.943	Weak baseline
Decision Tree	5-fold random	0.955	0.942	0.907	0.975	Moderate baseline
Random Forest (scikit-learn)	5-fold random	0.970	0.961	0.937	0.992	Strong baseline
ArcGIS Forest ★	Spatial CV (LODO)	0.933	0.869	0.830	0.973	Spatial CV

Table 8. Baseline model comparison

5.3 Variable Importance: SHAP Analysis

Figure 6 presents the bar chart and beeswarm plot together, while Figure 7 presents the dependence plot for distance to streams. Elevation's clear dominance in the model's decision-making (SHAP = 0.325, 49.7%) reveals that Ankara's topography is the most fundamental physical factor determining the distribution of urban flooding. This finding is independently corroborated by Gül's (2025) Ankara study.

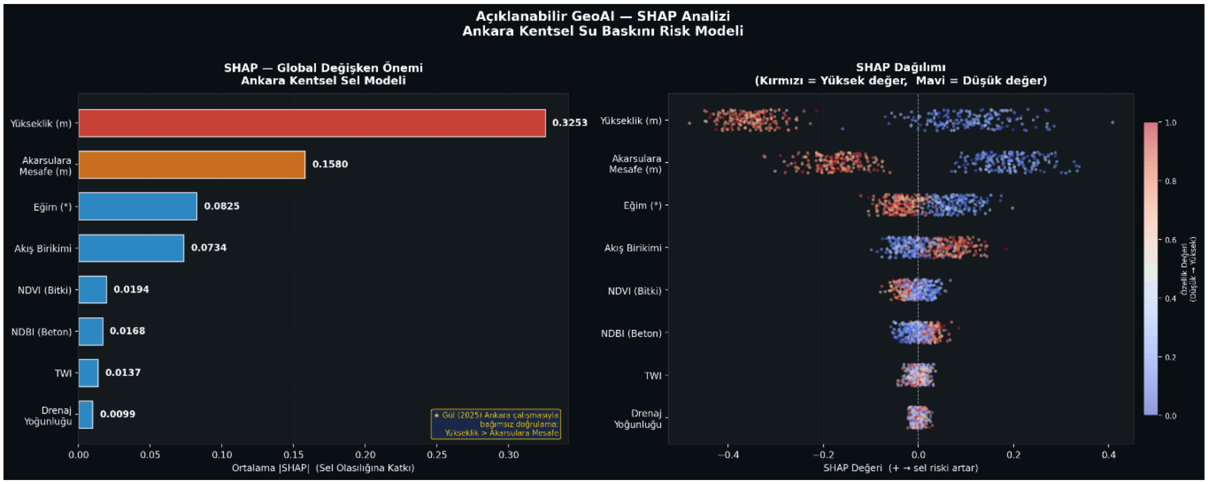


Figure 6. SHAP Analysis - Global Importance (left) and Beeswarm Distribution (right)

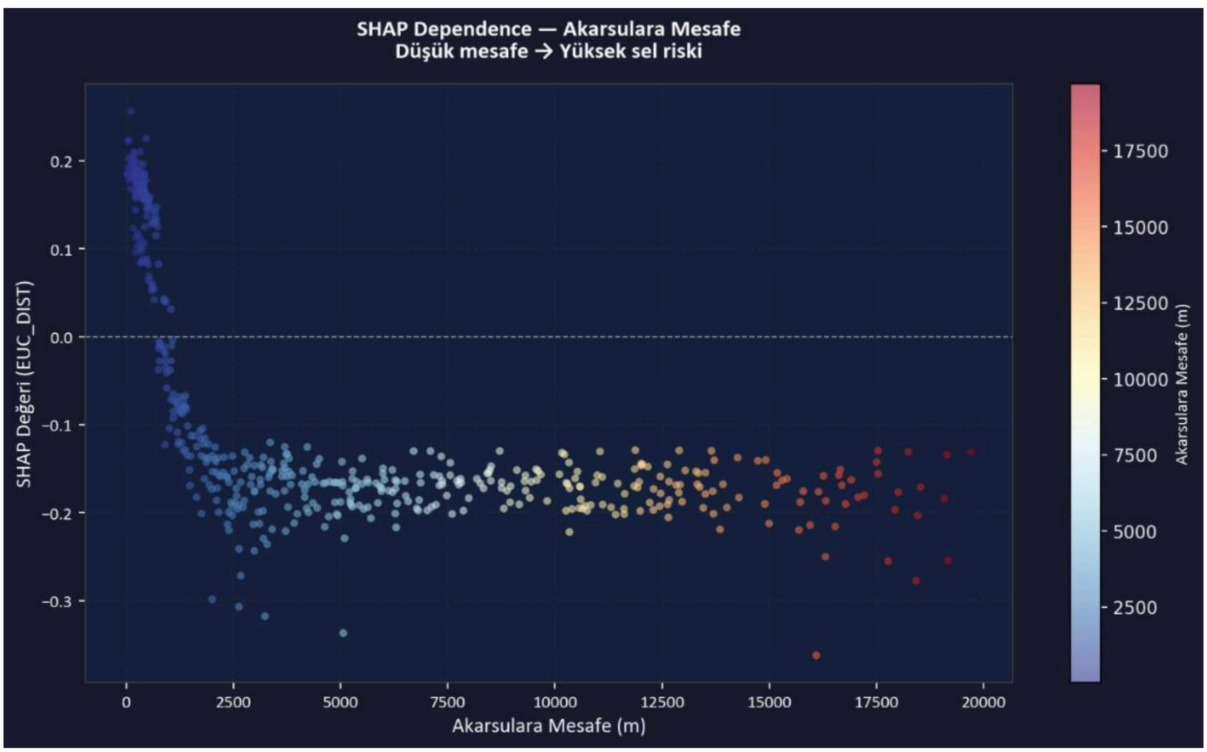


Figure 7. SHAP Dependence Plot - Distance to Streams (EUC_DIST)

Variable	SHAP Value	Relative Share (%)	Lit. Rank
Elevation (m)	0.3253	49.7%	1st
Distance to Streams (m)	0.1580	24.1%	5th
Slope (°)	0.0825	12.6%	2nd
Flow Accumulation	0.0734	11.2%	8th
NDVI (Vegetation Cover)	0.0194	3.0%	12th
NDBI (Built-up/Concrete)	0.0168	2.6%	-
TWI	0.0137	2.1%	6th
Drainage Density	0.0099	1.5%	9th

Table 9. SHAP global variable importance ranking

5.4 Temporal Trend Analysis

Figure 8 presents the Emerging Hot Spot map (Gi* statistic) and temporal trend, while Figure 9 presents the Cluster and Outlier Analysis (Anselin Local Moran's I). The 13 New Hot Spots detected during the 2020-2022 period are concentrated along the Etimesgut-Yenimahalle border and in Çankaya's southwestern corridor.

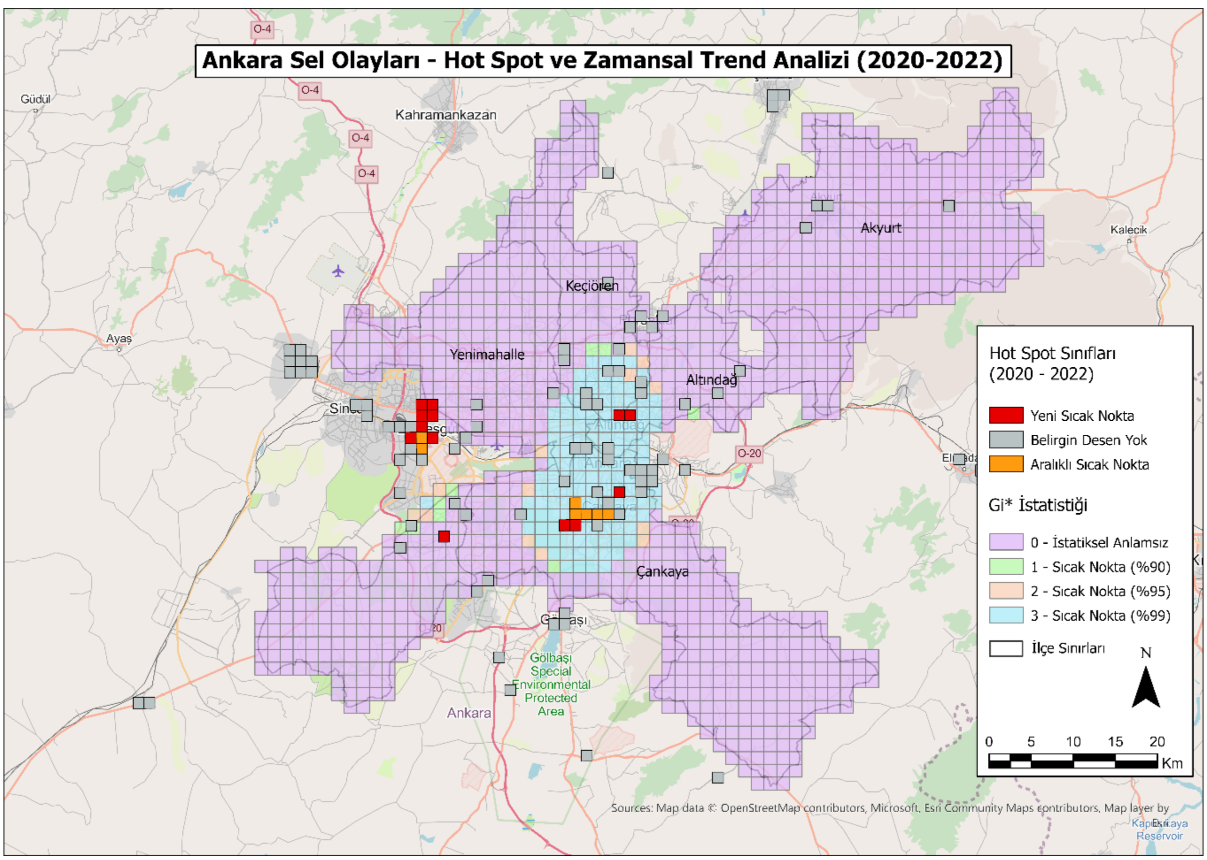


Figure 8. Ankara Flood Hot Spots and Temporal Trend Analysis (Space-Time Cube + EHS)

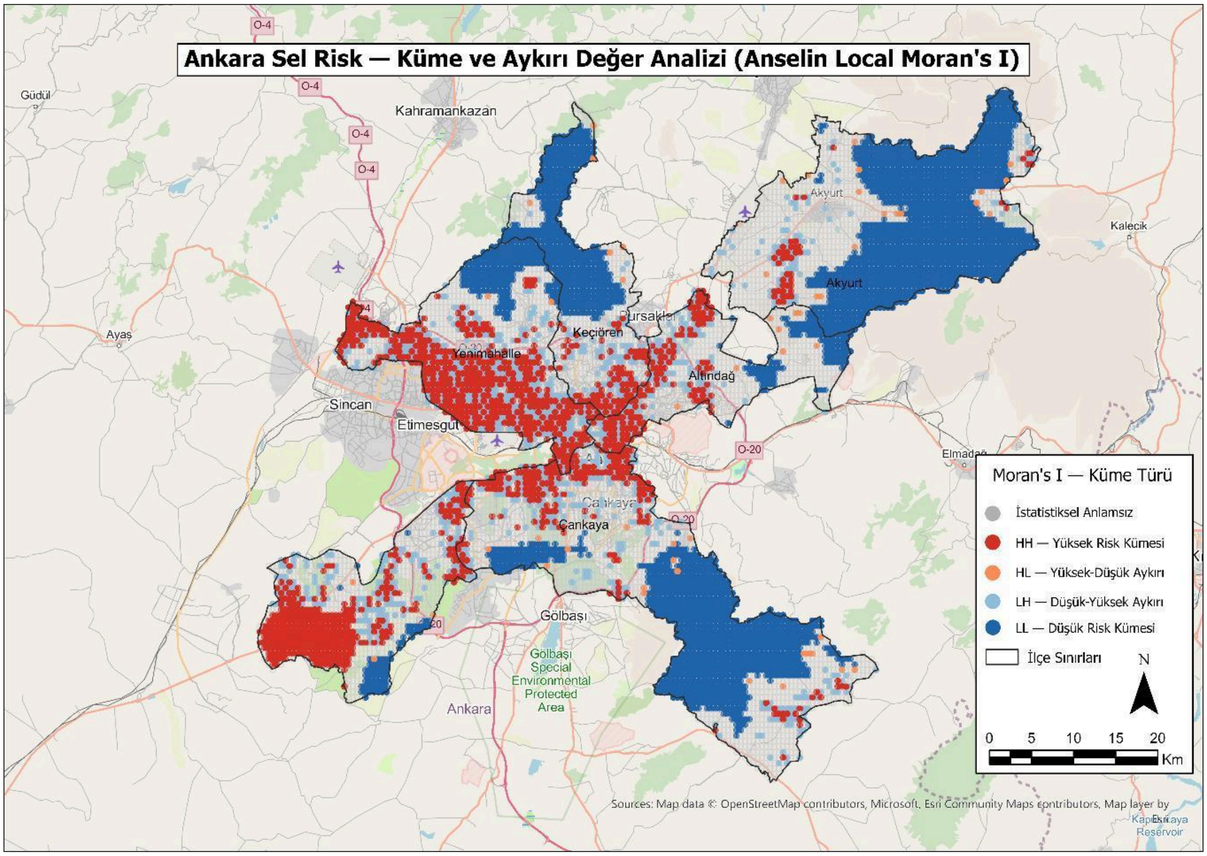


Figure 9. Cluster and Outlier Analysis - Anselin Local Moran's I

Pattern Type	Number of Areas	Description
New Hot Spot	13	Risk areas that have recently emerged
Sporadic Hot Spot	7	Areas showing periodic risk
No Pattern Detected	115	No clear trend
TOTAL	135	STC grid cells

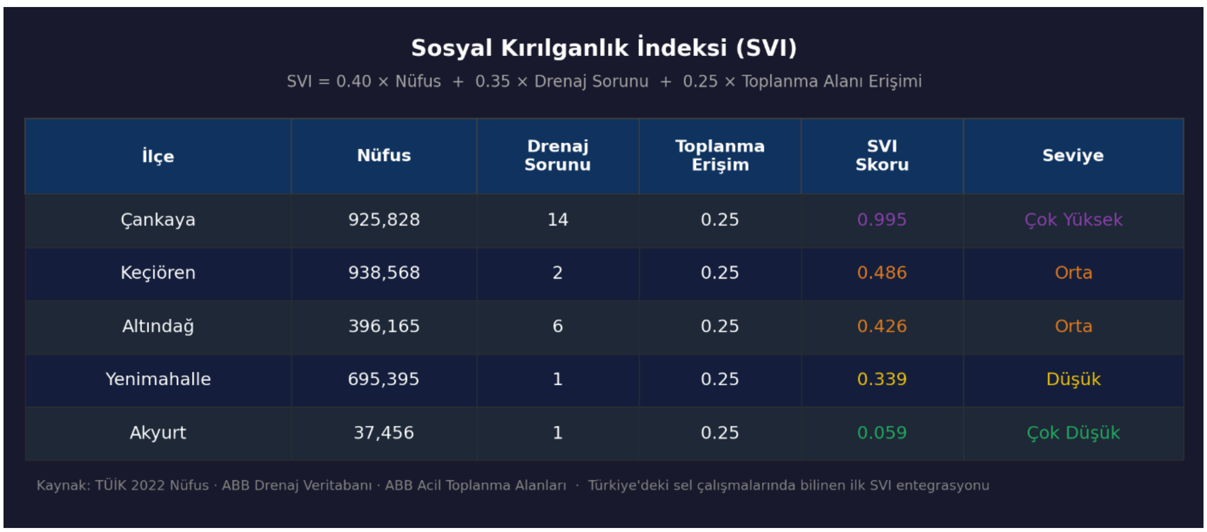
Table 10. Emerging Hot Spot Analysis pattern distribution

5.5 Population Risk Analysis and Social Vulnerability

Population in high-risk zones was calculated using the dasymetric proxy method. Table 11 presents the current situation and climate-scenario projections for 2035. Across the five districts, 666,009 people (22.1%) are under high flood risk. The SVI results are visualized in Figure 11, the zonal statistics in Figure 12, and emergency assembly-area accessibility in Figure 13.

District	Total Population	Under Risk	Share (%)	RCP 4.5 (2035)	RCP 8.5 (2035)
Keçiören	938,568	152,184	16.5%	167,065	175,928
Çankaya	925,828	181,818	19.4%	199,525	210,081
Yenimahalle	695,395	216,288	30.9%	237,431	249,918
Altındağ	396,165	110,782	26.9%	121,588	127,999
Akyurt	37,456	4,937	11.2%	5,420	5,707
TOTAL	3,019,904	666,009	22.1%	730,429	769,633

Table 11. Population under flood risk - current status and 2035 climate projections


Sosyal Kırılganlık İndeksi (SVI)

SVI = 0.40 × Nüfus + 0.35 × Drenaj Sorunu + 0.25 × Toplanma Alanı Erişimi

İlçe	Nüfus	Drenaj Sorunu	Toplanma Erişim	SVI Skoru	Seviye
Çankaya	925,828	14	0.25	0.995	Çok Yüksek
Keçiören	938,568	2	0.25	0.486	Orta
Altındağ	396,165	6	0.25	0.426	Orta
Yenimahalle	695,395	1	0.25	0.339	Düşük
Akyurt	37,456	1	0.25	0.059	Çok Düşük

Kaynak: TÜİK 2022 Nüfus · ABB Drenaj Veritabanı · ABB Acil Toplanma Alanları · Türkiye'deki sel çalışmalarında bilinen ilk SVI entegrasyonu

Figure 11. Social Vulnerability Index (SVI) - Comparison Across 5 Districts (source graphic labels in Turkish; translated data below)

District	Population	Drainage Issues	Assembly Access	SVI Score	Level
Çankaya	925,828	14	0.25	0.995	Very High
Keçiören	938,568	2	0.25	0.486	Medium
Altındağ	396,165	6	0.25	0.426	Medium
Yenimahalle	695,395	1	0.25	0.339	Low
Akyurt	37,456	1	0.25	0.059	Very Low

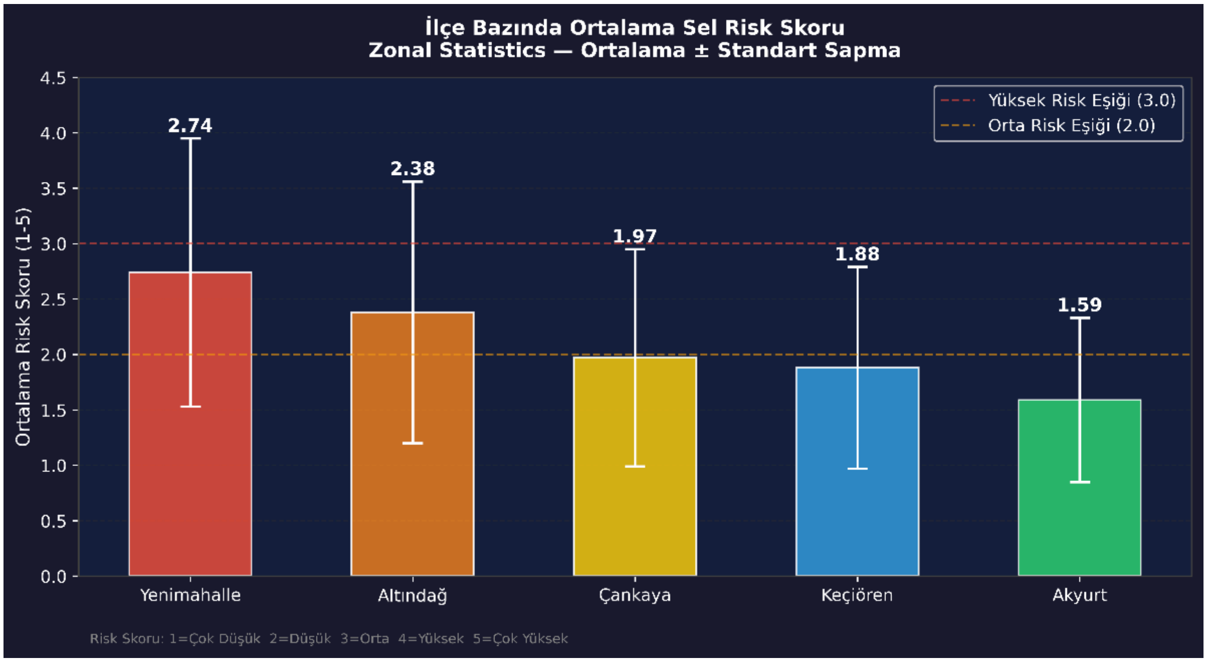


Figure 12. Average Flood Risk Score by District - Zonal Statistics

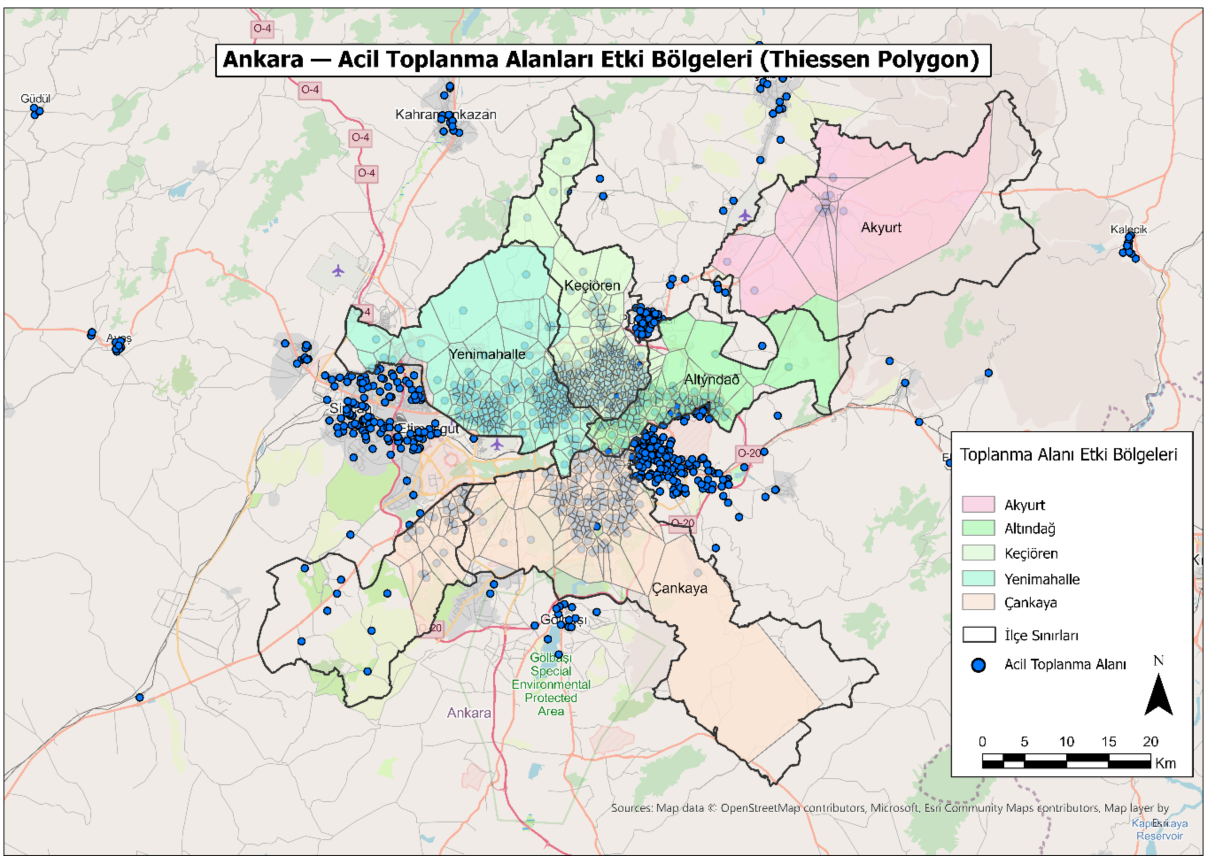


Figure 13. Emergency Assembly Area Service Areas - Thiessen Polygon Analysis

6. DISCUSSION

6.1 Methodological Originality

Dependent Variable Selection:

The ABB urban pluvial flood-report data represents events arising from insufficient infrastructure capacity and directly informs urban risk-management decisions. This type of data is a first for Turkey.

LODO vs. Random CV:

Random k-fold CV produced an MCC of 0.910, while LODO produced an MCC of 0.830 - concrete evidence of the +0.08-point artificial inflation caused by spatial autocorrelation.

Literature Corroboration via SHAP:

The ranking of Elevation (49.7%) above Distance to Streams (24.1%) is consistent with Gül (2025) and with the meta-analysis of Kaya & Derin (2023). This independent corroboration supports the model's reliability.

SVI Integration:

This study is the first example to carry out social-vulnerability integration in the context of urban flood studies in Turkey.

6.2 Limitations

- The training data spans a three-year period. A longer historical series could better capture seasonal variability.
- The SHAP analysis was carried out using scikit-learn; slight differences may exist relative to ArcGIS Forest.
- The five risk-class thresholds were determined using the equal-interval method; no hydrodynamic validation was performed with HEC-RAS.
- The SVI is limited to three components; PCA-based approaches could produce more comprehensive results.

7. CONCLUSION AND RECOMMENDATIONS

This study has produced a reliable and explainable GeoAI-based urban flood decision-support system by processing the Ankara Metropolitan Municipality's real urban pluvial flood-report data within a spatially honest machine-learning framework.

- LODO Spatial CV: MCC = 0.830, AUC = 0.973
- Independent validation (Akyurt, n=76): Recall = 93.4%, Precision = 100%
- SHAP: Elevation primary (49.7%), Distance to Streams secondary (24.1%) - corroborated by Gül (2025)
- 13 New Hot Spots - risk foci are shifting from the northwest toward the south
- 666,009 people (22.1%) under risk; RCP 8.5/2035: 841,248 people

Policy recommendations:

- The high-risk areas along the Çankaya-Keçiören-Altındağ axis should be a priority target for drainage rehabilitation.
- Building permits in the 13 New Hot Spot areas should be re-evaluated on a risk-focused basis.
- Emergency assembly-area accessibility should be improved in Çankaya, which has the highest SVI value (0.995).
- This study should be transferred to the ArcGIS Dashboard and StoryMaps platforms and converted into an interactive decision-support tool.
- Zoning plans for new construction areas should be updated based on the RCP 8.5 scenario.

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